

Typed Evaluation Models and the Collapse Between Conflict and Ignorance: A Case Study on Jev

Modelos de evaluación tipada y el colapso entre conflicto e ignorancia: un estudio de caso sobre Jev

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Abstract. Chat-based LLMs return free text that must be interpreted; a category of tools that emerged in September 2026, typed evaluation models ("System One models"), instead return a typed object directly usable by software. Jev (TypeSafe AI, reported by the vendor as launched on September 15, 2026) is the first commercial example of this category. This work situates it within a five-category landscape of AI model interaction and tests, with a minimal experiment, a prediction derived from annotated decision theory: that an output collapsed to a single calibrated probability cannot distinguish genuine conflict from genuine ignorance. For Jev's simplest question type (boolean/Noul), the experiment is consistent with that prediction: conflict cases (probability 0.50-0.57) and ignorance cases (0.46-0.48) fall within the same narrow band. But a second experiment, using the same model's Choice type with a schema that explicitly names "conflicting evidence" and "insufficient evidence" as options, separates both cases with probability 1.0 in all four completed cases. A third experiment, with Choice restricted to the two original options (no escape categories), reproduces neither the collapse nor the clean separation: an unanticipated directional bias appears, possibly tied to lexical cues in the input text. The central finding is therefore not that Jev collapses per se, but that the same model preserves, destroys, or unpredictably distorts the distinction depending on the question type and declared schema — an interface design risk more complex than a single follow-up experiment could anticipate.

Keywords: typed evaluation models; System One models; LLM-as-judge; annotated paraconsistent logic; representational collapse; typed abstention; Jev.

Resumen. Los modelos de chat basados en LLM devuelven texto libre que debe interpretarse; una categoría de herramientas que emergió en septiembre de 2026, los modelos de evaluación tipada ("System One models"), devuelven en su lugar un objeto tipado directamente utilizable por software. Jev (TypeSafe AI, reportado por el fabricante como lanzado el 15 de septiembre de 2026) es el primer ejemplo comercial de esta categoría. Este trabajo lo ubica dentro de un panorama de cinco categorías de interacción con modelos de IA y examina, con un experimento mínimo, una predicción derivada de la teoría de decisión anotada: que una salida colapsada a una sola probabilidad calibrada no puede distinguir conflicto genuino de ignorancia genuina. Para el tipo de pregunta más simple de Jev (boolean/Noul), el experimento es consistente con esa predicción. Un segundo experimento, con el tipo Choice y un esquema enriquecido, separa ambos casos con probabilidad 1,0. Un tercer experimento, con Choice restringido a las opciones originales, no reproduce ni el colapso ni la separación perfecta: aparece un sesgo direccional no anticipado. El hallazgo central no es que Jev colapse per se, sino que el mismo modelo preserva, destruye, o distorsiona la distinción según el tipo de pregunta y el esquema declarado.

Palabras clave: modelos de evaluación tipada; System One models; LLM-as-judge; lógica paraconsistente anotada; colapso representacional; abstención tipada; Jev.

1 Introduction

A conventional chat model, faced with the question "was the refund approved?", might answer "Yes, it seems so, although I'm not entirely sure..." — an unstructured string of text that the consuming software must re-parse or interpret, with no explicit score and no format guarantee. A category of tools that emerged in September 2026, typed evaluation models, inverts that relationship: the model is given a state (the available context or evidence) and a schema of questions, each with a declared type, and the model returns, for each question, a value of that type together with a calibrated probability.

TypeSafe AI calls its first model of this kind "System One" — the fast, intuitive counterpart to the slow, deliberate reasoning ("System Two") pursued by frontier LLMs [1]. According to the vendor, Jev does not generate text: it selects among the options of a schema supplied by the caller, so returning a malformed value or one outside the declared set would be structurally impossible [2] — a format guarantee, not a guarantee of content correctness. The vendor reports 20 to 200 times more speed and up to 400 times lower cost than a frontier LLM for this class of tasks [3], and a proprietary training method, Reinforcement Learning for Calibrated Decisions

(RLCD), aimed at explicit calibration [4]. None of these figures were independently audited in this work; they are reported as vendor claims, not as verified facts.

This work does not evaluate Jev's performance on its intended tasks. Before examining it as a case study (§3-§6), §2 situates it within a broader landscape of AI model interaction categories. A representational question is then examined — what does a model that collapses evidence to a single calibrated probability lose? — and, following an unexpected finding in the experiment itself, that question is refined into a more precise one: is the loss a property of the model, or of the type of interface chosen to query it?

2 Landscape of AI Model Interaction Categories

The binary opposition "free chat vs. typed" oversimplifies the real landscape. Figure 1 positions five categories along two axes: computational cost and rigidity/timing of the decision schema's fixation — not "input flexibility," a formulation from an earlier version of this figure that led to a misreading (it suggested that a generative chat model was less flexible than a typed model, when it is exactly the opposite in terms of what it can be asked). The correct axis runs from a schema fixed at training time (category 4, the most rigid) to a schema declared at each call (categories 2, 3, and 5), to the total absence of a schema (category 1, free text — the maximum possible expressive freedom, but with zero structural guarantee on the response).

1. Generative chat models — free text, no guarantee of format or type. This is the no-schema extreme of the axis: it accepts any question, but offers no guarantee about the shape of the answer.

2. LLM with forced output (constrained decoding) — still a general-purpose generative model underneath, but at each generation step the space of valid tokens is restricted according to a declared grammar or JSON schema [12]. The cost remains that of a full LLM.

3. LLM-as-judge — a frontier or fine-tuned LLM evaluates according to a natural-language criterion. When fine-tuned to a specific task, the literature documents that it can degenerate into a closed-domain classifier, with a performance drop outside that domain [13].

4. Classical discriminative classifiers — dedicated non-generative architectures (e.g., a fine-tuned BERT, or an evidential head such as EDL), deterministic, with a class schema fixed at training time. Reported as substantially cheaper and faster than an LLM-based judge, although the exact figure varies by source and is not taken here as precise data.

5. Jev and its peers (typed evaluation models) — the case study of this work.

Note on this taxonomy: it is not a strict partition. Categories 2 and 3 can overlap in practice — an LLM-based judge can use forced output to guarantee a valid JSON verdict — and are presented as heuristic axes (use, decoding technique, architecture, cost) rather than as mutually exclusive classes. It is the author's own synthesis of secondary sources, not a systematic review of the constrained-decoding and LLM-as-judge literature.

With that caveat, Jev sits at the convergence of categories 3 and 4: it inherits from category 3 the flexibility of a question schema declared at call time, and from category 4 the low cost and the guarantee of typed output by construction. The LLM evaluation ecosystem is converging toward two-tier architectures in 2026 — a cheap classifier that resolves most cases and escalates to a frontier judge only when the case is ambiguous [14] — which is, structurally, the same cascade that a typed-abstention taxonomy already formalizes for decision-making under insufficient or disputed evidence (the author's own unpublished research-program synthesis; see note in §7).

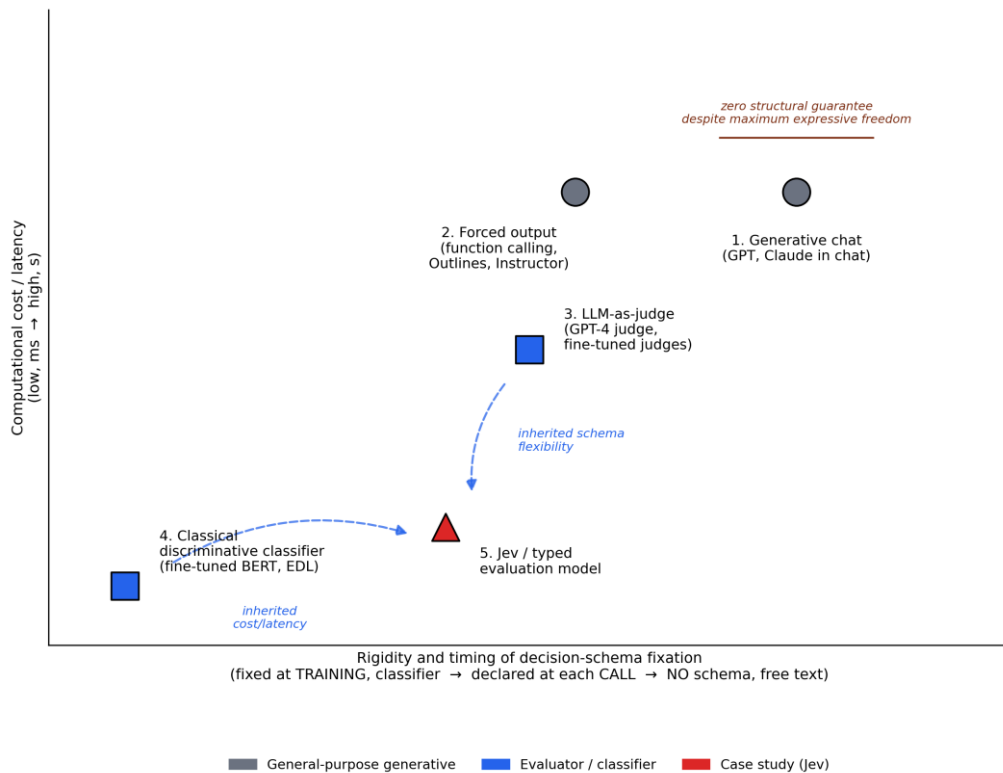


Figure 1: Positioning of the five AI model interaction categories according to schema rigidity/timing (x-axis: fixed at training → declared at each call → no schema) and computational cost (y-axis). Jev sits at the convergence between the LLM-based judge (category 3) and the classical discriminative classifier (category 4); generative chat (category 1) occupies the no-schema extreme — maximum expressive freedom, zero structural guarantee. Source: own elaboration.

3 Verified Taxonomy of Types in Jev

The vendor's documentation [8] specifies three question primitives, evaluable in parallel within a single call.

Table 1: Question primitives supported by Jev.

Type	What it does	What it returns
Choice	pick one option from a declared set (criteria: option→description map)	chosen option + probability per option
Score	rate the state on a rubric of ordered levels	score + probability per level
Noul	yes/no question	the probability that the answer is "yes"

Source: own elaboration.

The documentation calls the third type "Noul," not "Boolean," and describes it as a continuous probability in $[0,1]$ — not a discrete value. The Vercel AI Gateway SDK, which exposes `experimental_evaluate` as the access interface, names that same type `boolean` in its API signature [9]. Verified directly against the response object (§5): the field `providerMetadata.typesafe.confidence` exists in the schema but arrives empty (`{}`) in all five cases run with the boolean type — there is, for that type, no additional confidence or evidence-composition field beyond the single probability.

4 The Representational Question

In the annotated paraconsistent logic framework (LPA2v) [10], a judgment is represented as an independent pair (μ, λ) . Two distinguished vertices of the resulting lattice, $\top$ (inconsistent: strong evidence in both directions) and $\perp$ (paracomplete: absence of evidence in either direction), are epistemologically opposite but share a geometric property: both are equally far from the "true" and "false" ideal vertices [11]. From this a result already formalized follows: every closeness score to those ideal vertices is invariant under the involution that swaps $\top$ and $\perp$ [11].

This work does not apply that theorem literally to Jev — doing so would require Jev to explicitly compute an independent pair (μ, λ) , which is undocumented. The connection is a motivating analogy, not a deductive consequence of the theorem. The argument that is deductively valid, and that this work puts to the test, is simpler: no map from an evidence space of two or more dimensions (support, opposition, magnitude, insufficiency) to a sin-

gle scalar in $[0,1]$ can be injective; two distinct evidence states can, by pure cardinality, collapse to the same number. A model that exposes only that scalar — like Jev's Noul type — cannot, in principle, guarantee that conflict and ignorance produce distinct values.

5 Method

Six states (state) were designed in English on the same domain fact (whether a traffic light was red), in four categories: TORN (genuine conflict, $n=2$, two equally credible sources that contradict each other), SILENT (genuine ignorance, $n=2$, documented absence of evidence), AGREE-SUPPORT (control, $n=1$, sources agreeing on "red"), and AGREE-REFUTE (control, $n=1$, sources agreeing on "not red").

Experiment 1 (boolean/Noul type): one call per state to typesafe-ai/jev via experimental_evaluate from the ai SDK v7.0.107, single question: "Was the traffic light red at the time in question?"

Experiment 2 (Choice type, follow-up): the same six states, a single lightState question of type choice with four explicit options: red, green, conflicting_evidence ("reliable sources disagree with each other about the color"), and insufficient_evidence ("there is no evidence available to determine the color"). This second experiment was designed after observing the results of Experiment 1, prompted by an adversarial review that flagged the need to test whether the collapse was a property of the model or of the question type — it is explicitly declared as not preregistered and data-motivated.

Experiment 3 (binary Choice, closing the open question): the same six states, the same lightState question of type choice, but with only the two original options: red and green, no escape categories. Tests whether the perfect separation of Experiment 2 depends on the schema being able to name conflict/insufficiency, or whether it is a general property of the Choice type independent of the schema. Also data-motivated, not preregistered.

None of the three experiments averages repetitions (a single call per case; limitation in §8). The calls were made on September 19, 2026 against the production model, with no access to its weights or training set.

6 Results

Table 2: Experiment 1 (boolean/Noul type) — probability of "red".

Case	Category	P(red)
TORN-1	genuine conflict (contradictory witnesses)	0.57
TORN-2	genuine conflict (contradictory sensors)	0.50
SILENT-1	genuine ignorance (no witnesses or cameras)	0.46
SILENT-2	genuine ignorance (lost record)	0.48
AGREE-SUPPORT	control: sources agree (support)	0.83
AGREE-REFUTE	control: sources agree (refutation)	no data (rate-limit, §8)

Source: own elaboration.

The two TORN cases (0.50–0.57) and the two SILENT cases (0.46–0.48) fall within a common, narrow band. The only completed control with unambiguous evidence (AGREE-SUPPORT) yields 0.83, far from 0.5.

Table 3: Experiment 2 (Choice type, enriched schema) — choice and probability.

Case	Category	Jev's choice	P(choice)
TORN-1	genuine conflict	conflicting_evidence	1.00
TORN-2	genuine conflict	conflicting_evidence	1.00
SILENT-1	genuine ignorance	insufficient_evidence	1.00
SILENT-2	genuine ignorance	insufficient_evidence	1.00
AGREE-SUPPORT	control: sources agree	red	1.00
AGREE-REFUTE	control: sources agree	no data (rate-limit, §8)	—

Source: own elaboration.

With the Choice type and a schema that explicitly names "conflicting evidence" and "insufficient evidence" as first-class options, Jev separates the four experimental cases (TORN vs. SILENT) with maximum probability and no ambiguity, and correctly classifies the support control. The same textual evidence that collapsed to a 0.46-0.57 band under the boolean type produces a perfect distinction under the Choice type.

Table 4: Experiment 3 (binary Choice type, red/green, no escape categories).

Case	Category	P(red)	P(green)
TORN-1	genuine conflict	0.85	0.15
TORN-2	genuine conflict	0.84	0.16
SILENT-1	genuine ignorance	0.67	0.33
SILENT-2	genuine ignorance	0.77	0.23

AGREE-SUPPORT	control: sources agree	1.00	0.00
AGREE-REFUTE	control: sources agree	no data (rate-limit, §8)	—

Source: own elaboration.

Without escape categories, Jev does not reproduce the ~ 0.5 collapse of Experiment 1 nor the perfect separation of Experiment 2. Instead, a directional bias toward red appears across all six cases, more pronounced in TORN (0.84-0.85) than in SILENT (0.67-0.77) — a weak ordinal trend ($n=2$ per category) that does not allow a firm conclusion. It is a third pattern, unanticipated by either of the two simple hypotheses (symmetric collapse or clean separation).

7 Discussion

The pattern in Experiment 1 is consistent with the prediction in §4: under the Noul type, Jev does not distinguish why the evidence is unclear. But Experiment 2 forces a correction of the interpretation: the loss is not a property of the Jev model, but of the question type and schema declared to query it. The boolean type forces, by definition, an output in a one-degree-of-freedom space — there is no way for it to return "conflict" or "insufficiency" even if the model internally distinguishes them. The Choice type, when the schema includes those categories as named options, does have representational room to express them, and the model exploits it with a sharpness that was not anticipated (probability 1.0, not a diffuse tendency).

This reframes the contribution of the work: it is not "Jev has an inherent representational limitation," but rather that the design of the query schema determines whether the distinction survives, and the simplest, cheapest type to use (Noul, a single yes/no question) is precisely the one that destroys it by construction. This is a usage risk, not a product engineering defect.

Experiment 3 answers the open question from v0.2 only partially, and in a more uncomfortable way than either of the two simple answers that were anticipated. The hypothesis "Choice always separates, regardless of the schema" is ruled out: without escape categories, there is no clean separation (0.84 is not 1.00, and SILENT does not approach 0.5 from the other side). But the alternative hypothesis "without escape categories, Choice collapses just like Noul" is also ruled out: there is no symmetric collapse toward 0.5 in any case — all values, including the SILENT cases with no evidence whatsoever, lean toward red.

The most honest explanation available with this data, without over-interpreting an $n=2$ per cell, is a possible lexical confound in the design of the stimulus itself: the TORN and AGREE-SUPPORT states literally contain the word "red" in the text (someone asserts it, even if it is contested), while the SILENT states do not mention either "red" or "green" at all. If the model partly weighs the surface presence of the word naming an option — and not only whether that assertion is credible or disputed — that would produce exactly the observed pattern. This explanation is plausible and testable, but this work did not test it in a controlled way — it remains a genuine open question, not a closed finding.

What does hold up across the three experiments together: the deductive argument in §4 (no single scalar can be injective over a higher-dimensional evidence space) explains the collapse of the Noul type, but does not predict or explain the pattern of Experiment 3 — a Choice with two options is not a single injective scalar in the same way, and yet it failed to achieve the separation that its representational space would in principle allow. The practical recommendation stands, but becomes more cautious: to preserve the distinction between conflict and ignorance, avoiding the boolean type is not enough — the Choice schema must also explicitly name those states as first-class options, and it is worth empirically verifying that the model is not responding to superficial lexical cues in the state rather than to the actual structure of the evidence.

8 Limitations

(a) $n=5$ completed cases out of 6 in each of the three experiments (the AGREE-REFUTE control failed due to the Vercel AI Gateway free tier's rate limit in all three, even with a payment method already on file — the provider requires loaded paid credits, not just a card on file, and that step was not completed; it is the only case that could never be observed, under any condition); (b) a single call per case in each experiment, with no repetition to estimate variance — critical for Experiment 3, whose pattern ($n=2$ per category) does not allow distinguishing a real trend from sample noise; (c) Experiments 2 and 3 were motivated by prior results and were not preregistered — explicitly declared as such; (d) the lexical-bias explanation for Experiment 3 (§7) is plausible but was not tested in a controlled way; (e) this is a four-day-old product at the time of writing — its behavior and limits may change without notice, and these results should be read as a snapshot of 2026-09-19; (f) the speed, cost, and training-method (RLCD) figures are vendor claims, not independently audited in this work; (g) the five-category taxonomy in §2 is the author's own synthesis, not a systematic review.

9 Conclusion

The first experiment in this work seemed to confirm that typed evaluation models inherit, by representational design, the collapse between conflict and ignorance already formally characterized in annotated decision theory.

A second experiment showed that conclusion to be incomplete: the same model, with a Choice schema that explicitly names the disputed epistemic states, separates conflict from ignorance with maximum probability. A third experiment, designed to isolate whether that separation depended on the schema or on the question type, showed that reality is messier than either of those two stories: without escape categories, Jev neither collapses symmetrically nor separates cleanly, but instead exhibits an unanticipated directional bias, possibly tied to superficial lexical cues in the input text.

The lesson that survives all three experiments is not about a fixed limit of Jev as a product, but about two distinct and equally real risks for anyone building on typed evaluation models: first, reducing an epistemically complex decision to the simplest available interface type (boolean/Noul) discards information by construction; second, even a more expressive interface (Choice) can fail in non-obvious ways — biased, not simply "less informative" — when the schema does not explicitly name the states that matter.

Conflict of Interest

The author declares no conflict of interest.

Data and Code Availability

Public repository: <https://github.com/mleyvaz/jev-typed-evaluation-collapse> — includes the script and results of Experiment 1 (run_experiment.mjs, results.json), Experiment 2 (run_experiment_choice.mjs, results_choice.json), Experiment 3 (run_experiment_choice_binary.mjs, results_choice_binary.json), the Figure 1 script (make_fig1_taxonomy.py), and the adversarial review round that motivated the v0.1 → v0.2 → v0.3 revisions.

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Appendix A. Input States (verbatim)

The six states (state) used across the three experiments, identical in each — only the declared question (questions) changes.

TORN-1 (genuine conflict)

Witness A, a police officer with a clear view of the intersection, testified under oath that the traffic light was red at the moment of the collision. Witness B, a bystander standing next to Witness A with an equally clear view, testified under oath that the same traffic light was green at that exact moment. Both witnesses are considered reliable by the investigating officer; there is no indication either one is lying.

TORN-2 (genuine conflict)

Sensor 1, a calibrated high-precision traffic sensor, recorded the light as RED at 14:03:02. Sensor 2, an equally calibrated high-precision sensor mounted on the same pole, recorded the light as GREEN at the same timestamp, 14:03:02. Both sensors passed their most recent calibration check with no faults reported.

SILENT-1 (genuine ignorance)

No witnesses were present at the intersection at the time in question. No traffic cameras were operating in that area that day. There is no record, sensor log, or testimony of any kind describing the state of the traffic light at that moment.

SILENT-2 (genuine ignorance)

The traffic light's camera log for that day was permanently lost in a server failure before any backup was made. No witnesses have come forward. No other observation of the light exists in any form.

AGREE-SUPPORT (control, support)

Witness A testified that the traffic light was red at the time of the collision. Witness B, standing nearby with an independent line of sight, separately and independently confirmed that the light was red at that same moment.

AGREE-REFUTE (control, refutation)

Witness A testified that the traffic light was green, not red, at the time of the collision. Witness B, standing nearby with an independent line of sight, separately and independently confirmed that the light was green at that same moment.

Appendix B. Call Code, per Experiment

Experiment 1 (boolean/Noul):

```
import { experimental_evaluate as evaluate } from 'ai';

const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT, // one of the six states in Appendix A
  questions: {
    wasRed: {
      type: 'boolean',
      instructions: 'Was the traffic light red at the time in question?',
    },
  },
});
```

Experiment 2 (Choice, enriched schema):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: {
    lightState: {
      type: 'choice',
      instructions: 'What was the state of the traffic light at the time in question?',
      criteria: {
        red: 'The evidence indicates the light was red.',
        green: 'The evidence indicates the light was green (not red).',
        conflicting_evidence: 'Reliable sources disagree with each other about the color.',
        insufficient_evidence: 'There is no evidence available to determine the color.',
      },
    },
  },
});
```

Experiment 3 (binary Choice, no escape categories):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: {
    lightState: {
      type: 'choice',
      instructions: 'What was the state of the traffic light at the time in question?',
      criteria: {
        red: 'The evidence indicates the light was red.',
        green: 'The evidence indicates the light was green (not red).',
      },
    },
  },
});
```

Appendix C. Full Response (answers) per Case

The answers object returned by Jev for each completed case, unedited (the full response object also includes usage, warnings, rounding, and providerMetadata, available in the results*.json files in the repository).

Experiment 1 — boolean/Noul:

```
TORN-1:      {"wasRed":{"type":"boolean","probability":0.57}}
TORN-2:      {"wasRed":{"type":"boolean","probability":0.50}}
SILENT-1:    {"wasRed":{"type":"boolean","probability":0.46}}
SILENT-2:    {"wasRed":{"type":"boolean","probability":0.48}}
AGREE-SUPPORT: {"wasRed":{"type":"boolean","probability":0.83}}
AGREE-REFUTE: ERROR - GatewayRateLimitError (free-tier rate limit, S8)
```

Experiment 2 — enriched Choice:

```
TORN-1:      {"lightState":{"choice":"conflicting_evidence","probabilities":
               {"red":0,"green":0,"conflicting_evidence":1,"insufficient_evidence":0}}}
TORN-2:      {"lightState":{"choice":"conflicting_evidence","probabilities":
               {"red":0,"green":0,"conflicting_evidence":1,"insufficient_evidence":0}}}
SILENT-1:    {"lightState":{"choice":"insufficient_evidence","probabilities":
               {"red":0,"green":0,"conflicting_evidence":0,"insufficient_evidence":1}}}
SILENT-2:    {"lightState":{"choice":"insufficient_evidence","probabilities":
               {"red":0,"green":0,"conflicting_evidence":0,"insufficient_evidence":1}}}
AGREE-SUPPORT: {"lightState":{"choice":"red","probabilities":
               {"red":1,"green":0,"conflicting_evidence":0,"insufficient_evidence":0}}}
AGREE-REFUTE: ERROR - GatewayRateLimitError (free-tier rate limit, S8)
```

Experiment 3 — binary Choice:

```
TORN-1:      {"lightState":{"choice":"red","probabilities":{"red":0.85,"green":0.15}}}
TORN-2:      {"lightState":{"choice":"red","probabilities":{"red":0.84,"green":0.16}}}
SILENT-1:    {"lightState":{"choice":"red","probabilities":{"red":0.67,"green":0.33}}}
SILENT-2:    {"lightState":{"choice":"red","probabilities":{"red":0.77,"green":0.23}}}
AGREE-SUPPORT: {"lightState":{"choice":"red","probabilities":{"red":1,"green":0}}}
AGREE-REFUTE: ERROR - GatewayRateLimitError (free-tier rate limit, S8)
```